

Excess Separability: Nuisance-Controlled Residual-Stream Probing for Benchmark Contamination Detection

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Abstract

Benchmark contamination is diagnosed today with n -gram overlap, with likelihood-based membership inference, or with canary strings, and each needs something usually unavailable: the training corpus, a well-chosen test statistic, or foresight at dataset release. A recent alternative reads contamination off a linear probe on the model’s internal activations. We show that the natural way to do this does not work, and specify one that survives measurement.

The protocol reports a zero-sum contrast on the depth profile of probe accuracy, recentred on a level-matched placebo baseline and tested against a label-permutation null, with the reference set twice the size of the suspect set. Each of those four choices replaces a simpler alternative that we measured and rejected. Reporting the level of excess separability rather than its shape makes the false positive rate track the size of the analyst’s own control set, from 0.03 to 0.99 under a true null. Contrasting against a flat depth profile fails in both directions, rejecting a true null 0.72 of the time when surface decodability rises with depth and losing all power when it falls, which is what a residual stream with accumulating noise produces. An item bootstrap holds the fitted probe fixed and rejects up to 0.09 of the time where a permutation null that refits it holds 0.02. A half-size baseline triples the error rate. The corrected protocol holds the nominal rate across all of these, under two deliberately dissimilar simulators, and reaches 0.8 power at a separation of roughly one to two accuracy points between the embedding layer and the profile peak.

We then run the protocol on real transformers. Baseline depth profiles are measurably not flat, spanning up to 29.1 accuracy points on a temporal split, and their non-flatness tracks the surface difference between the two item sets (correlation 0.87 over 6 audits), so the correction is largest

exactly where it is needed and negligible where it is not. All 4 well-matched Pile arms return null, as the literature predicts for a corpus seen approximately once, and the protocol refuses a verdict on the temporal split rather than reporting one. What this does not establish is whether transformers carry a familiarity direction at all: the only positive sits on the split where exchangeability fails. We release the implementation, the test suite, the validation study and the audits.

1 Introduction

A benchmark score is a claim about capability. Contamination breaks the claim by making the score partly a report of what the model has memorised. The problem is old, was documented at scale in the GPT-3 report [10], and has grown worse as training corpora have grown and as their contents have stopped being disclosed [3, 54].

The detection tools available today divide into three families, each blocked by a different missing precondition. Substring and n -gram overlap search requires the training corpus, which for most released models does not exist publicly [22, 26]. Likelihood-based membership inference requires only the model, but fixes a scalar test statistic in advance (loss, minimum- k % token log-probability, a reference-model ratio) and inherits whatever that statistic fails to capture [46, 55, 68]; evaluated under splits that are genuinely exchangeable, this family performs close to chance on pretraining-scale data [24]. Canary insertion works well but has to be planned before the benchmark is published [13], which does nothing for the benchmarks already in use.

A fourth option has appeared recently. Liu et al. [42] train a linear probe on a model’s internal activations and report state-of-the-art AUC on WikiMIA and on a new arXiv-based benchmark. The motivation is sound: if memorisation leaves any trace in the model, it should be visible in the residual stream before it is visible in a hand-picked summary of the output distribution, and a probe learns the relevant direction instead of assuming it.

The difficulty is that a probe trained to separate two item sets will use anything that separates them. Das et al. [20] show that on several widely used membership-inference

Code, item-set builders, the Phase 0c validation study and the artefact manifest are at github.com/mabushi-lab/residual-stream-contamination-probing. The exact snapshot these results were produced from is archived at doi:10.5281/zenodo.21855510. Upstream dataset revisions are pinned, so the item sets rebuild to the hashes recorded in the manifest. The Pile extracts are rebuilt rather than redistributed; GSM1k is not redistributable at all.

benchmarks, including WikiMIA, a classifier with no access to the target model at all reaches high AUC, because the member and non-member sets were split by publication date and therefore differ in topic and style. Any probe evaluated on such a split reports the sum of two quantities, memorisation and distribution shift, and the split’s construction guarantees the second is large. The strongest evidence in Liu et al.’s own paper is consistent with this reading: on the deliberately contaminated checkpoints of Oren et al. [49], where the contaminated and clean item sets come from the same dataset and the split is not temporal, their probe reaches 54.0 and 51.9 AUC at duplication counts of 1 and 2, against a 50.0 chance baseline.

This paper takes the position that the probe is the right instrument and the reported quantity is the wrong one. We propose **Residual-Stream Contamination Probing (RSCP)**, a protocol whose output is the probe’s advantage *over a declared nuisance family*, tested against a permutation null, and calibrated against known exposure counts.

Contributions.

1. An exchangeability requirement on the reference set, stated as a precondition rather than a caveat, with three constructions that satisfy it and a diagnosis of which existing evaluation sets do not (§3.2).
2. An extraction rule: activations are read at the item prefix, because a state computed over the answer lets the detector read the model’s own competence, which is the effect contamination is supposed to explain (§3.3).
3. A statistic that is a contrast on the depth profile rather than a level, which cancels the analyst’s control set algebraically (§4, C1).
4. A level-matched placebo baseline, which removes the assumption that surface decodability is flat in depth. This is the paper’s main technical contribution: the same construction that vetoes an unusable audit also supplies the baseline that makes a usable one correct (§4, C2).
5. A permutation null on a label-free smoother, which refits the probe and so captures the variance an item bootstrap misses, at one matrix-vector product per draw (§4, C3).
6. A validation study under two structurally dissimilar simulators that falsified three of our own earlier design choices, together with the reasoning that replaced them (§4).
7. Phase 1 of the empirical programme, executed: 6 audits on real Pythia models establishing that baseline depth profiles are not flat, that their non-flatness scales with the surface difference, and that the protocol’s negative control behaves (§8).
8. A runnable implementation, a property test suite, and a build that regenerates every number in this paper from scratch in about ten minutes on a laptop (§9).
9. A pre-registered follow-on programme with falsification criteria and a compute budget on the order of 100 GPU-hours (§10).

One boundary is worth drawing at the outset. Section 4 reports measurements of a statistical procedure on synthetic activations; Section 8 reports measurements on real models. Neither is evidence that a transformer carries a familiarity direction, for reasons §8 sets out. Everywhere else, every number is either a published measurement with a citation or a synthetic value marked *illustrative* to show the shape of the output rather than to anticipate a result.

2 Background and Related Work

2.1 What contamination is, and what makes it hard to see

Contamination is not a single phenomenon. Verbatim inclusion of a test item in the pretraining corpus, inclusion of a paraphrase, inclusion of the item’s source document without the question-answer framing, and inclusion of a downstream discussion of the benchmark all inflate scores by different mechanisms and to different degrees. Magar and Schwartz [43] separate memorisation, meaning the model has stored the item, from exploitation, meaning the stored item actually changes the answer, and show the two come apart: a model can memorise contaminated items without exploiting them, and the size of the exploitation effect depends on the fine-tuning regime. Yang et al. [64] show that paraphrased benchmark items evade n -gram detection while still inflating scores, which rules out overlap search as a complete solution even when the corpus is available.

The corpus is usually not available, and where it is, overlap search has free parameters that matter: the GPT-3 analysis [10] and the C4 documentation study [22] use different n -gram lengths and normalisation and are not directly comparable. Corpus-search tooling has improved [26, 57], but only for models whose corpora are published.

Corpus-free methods therefore carry the field. Sainz et al. [54] prompt models to reproduce dataset splits verbatim. Golchin and Surdeanu [29] use guided instructions and a quiz format. Deng et al. [21] exploit the model’s ability to reconstruct masked incorrect options. Li and Flanigan [41] use release dates as a natural experiment. Chang et al. [17] infer which books a model has seen from cloze accuracy on passages, which is the same idea applied to copyrighted text rather than to benchmarks. Oren et al. [49] give the most statistically careful of these: benchmark items are

exchangeable under permutation of their order, so a model that assigns higher log-likelihood to the canonical ordering than to shuffled orderings has seen the canonical ordering, and this yields a valid p -value with no distributional assumptions. Its limitation is that it detects contamination of the dataset *as a file*, so it is blind to contamination that arrived through paraphrase, through a different serialisation, or one item at a time.

2.2 Membership inference and its exchangeability problem

Membership inference asks whether a specific example was in the training set [56, 65]. The strongest formulations calibrate against a reference distribution over models [15], which is unavailable at pretraining scale, so LLM-oriented work uses reference-free statistics: the loss itself [65], likelihood ratios adapted from the masked-language-model setting [47], comparison against synthetically perturbed neighbours [46], the mean log-probability of the lowest-probability $k\%$ of tokens [55], and a normalised variant motivated by the observation that training samples sit at local maxima of the modelled distribution along each input dimension [68].

Duan et al. [24] evaluate this family across Pythia models from 160M to 12B on Pile train versus Pile test and find performance barely above chance, attributing it to a very large corpus seen close to once and a member boundary made fuzzy by near-duplicates. Das et al. [20] sharpen the consequence: where the split is temporal, a blind classifier that never queries the target model achieves high AUC, so high AUC is not evidence of memorisation detection. Together these say that reported membership-inference performance on LLMs is largely an artefact of how the two sets were built, and that under exchangeable splits the true signal is small.

Maini et al. [44] draw the constructive conclusion. Individual membership is close to undetectable, but aggregating a weak per-item signal over a whole dataset is a different and much easier statistical problem. Their dataset inference procedure combines several per-item statistics and tests at the set level. RSCP sits in this tradition: it is a two-sample test on sets, not a classifier deployed on individual items, and we frame it that way throughout.

2.3 Probing classifiers: what a probe licenses

Linear probes read a property off intermediate representations [1], and the interpretive difficulty has been mapped carefully. Hewitt and Liang [34] show that probe accuracy confounds what the representation encodes with what the probe can learn, and introduce control tasks with randomised labels together with selectivity. Voita and Titov [62] and Pimentel et al. [50] recast probing information-theoretically and note that a sufficiently expressive probe recovers any property computable from the input, so accu-

racy alone is uninformative without a baseline. Belinkov [4] surveys the consensus: probe accuracy means something only relative to a declared control, and a negative result is weak evidence of absence, showing only that the property is not *linearly* decodable at the chosen extraction point.

RSCP inherits all of this. The controls we require, in §3.5, are the direct descendants of Hewitt and Liang’s control tasks, extended in one direction that their setting did not need: because the property under test here is defined by a difference between two item *sets*, the relevant control has two parts: a randomised-label task, and the strongest classifier buildable from the surface difference between the sets.

2.4 Why one might expect a familiarity direction

The mechanistic prior for RSCP is that models carry linearly decodable signals about their own epistemic state. Models have calibrated internal representations of whether they will answer correctly [37]. Truth of a declarative statement is linearly decodable from residual-stream activations across several datasets, with the recovered directions transferring between datasets [45], can be recovered without labels from consistency constraints [11], and separates statements the model itself is asserting falsely [2]. Sparse probing finds context and domain features represented in individual neurons [31]. Representation-engineering work shows that such directions are not merely correlational, since adding them to the residual stream changes behaviour in the predicted direction [53, 69].

None of this establishes that a *familiarity* direction exists. It establishes that the hypothesis is of a type that has been confirmed before, which is the right standing for a proposal paper. It also sets up the most informative possible negative result, discussed in §6: if excess separability is near zero on models with verified single-exposure contamination, then representational familiarity has an exposure floor, and the field should stop expecting activation-based detectors to catch low-duplication leakage.

Figure 1 places these families side by side. What it makes visible is that nothing in the existing set declares a nuisance control or reports contamination as a graded quantity rather than a verdict.

2.5 The immediate predecessor

Liu et al. [42] is the closest work and the one RSCP is designed to correct. Their pipeline fine-tunes the target model on a set of documents to produce a proxy model that has demonstrably memorised them, trains a logistic probe on last-token activations from the proxy at a validation-selected layer, and applies that probe to the original target model. On WikiMIA with Pythia-2.8B they report 69.8 AUC against 65.5 for the best baseline, and on their Arx-

	n-gram overlap	Likelihood MIA	Black-box order test	Activation probe (prior)	RSCP (this paper)
Works without the training corpus	●	●	●	●	●
Works without model weights	●	●	●	●	●
Needs no foresight at dataset release	●	●	●	●	●
Detects paraphrased contamination	●	●	●	●	●
Detects item-level, not just file-level	●	●	●	●	●
Reports a calibrated uncertainty	●	●	●	●	●
Declares a nuisance control	●	●	●	●	●
Reports an exposure level, not a verdict	●	●	●	●	●
	● yes	● partial	● no		

Figure 1: Preconditions and capabilities of the four existing families and of RSCP. No column dominates: the black-box order test is the only option for a closed model, and overlap search is the only one that says which documents leaked. The bottom two rows are what this paper adds, and the second row is what it gives up. Judgements, not measurements.

ivMIA benchmark with OpenLLaMA-13B they report 60.3 against 55.9.

Four features of that design are what RSCP changes. Both evaluation sets are split temporally: WikiMIA members are pre-cutoff Wikipedia event text, and ArxivMIA members come from the RedPajama arXiv subset against abstracts published after 2024. These are the splits Das et al. [20] show to be separable blind, so the reported AUCs upper-bound rather than estimate the memorisation signal. The extraction point is the last token of a template that wraps the full sample and appends an evaluative question, reported as “Here is a statement: [SAMPLE] Is the above statement correct? Answer:”. That state is downstream of the entire item and the prompt explicitly asks the model to evaluate it; for benchmark contamination the consequences are severe, because competence on the item is precisely what contamination inflates (§3.3). The layer is selected on a validation set and reported without correction for the search. And the probe transfers from a fine-tuned proxy to the target model, which assumes that the direction separating members after supervised fine-tuning is the direction pretraining memorisation writes to; that assumption is substantive and is not tested.

Their result on the deliberately contaminated checkpoints of Oren et al. [49], where the split is not temporal, is the honest calibration point for the whole approach and we return to it in §5.2.

3 Method

Figure 2 shows the whole procedure before the pieces are specified individually.

3.1 Setup and notation

Let M be a decoder-only transformer with L blocks and hidden width d . For a token sequence $x = (x_1, \dots, x_T)$,

write $h_t^{(\ell)}(x) \in \mathbb{R}^d$ for the residual stream at position t after block ℓ , with $h^{(0)}$ the embedding plus positional contribution and no attention or MLP applied.

A benchmark item is a pair (q, a) where q is everything the model is shown before it must produce an answer, including any few-shot context, instructions, and answer options, and a is the gold continuation. Write $p = q$ for the prefix and $f = q \oplus a$ for the full item.

We are given a *suspect set* $S = \{s_1, \dots, s_n\}$ of items drawn from the benchmark under audit and a *reference set* $R = \{r_1, \dots, r_n\}$ of items believed absent from the model’s training data. Assign $y_i = 1$ to items of S and $y_i = 0$ to items of R , giving a pooled sample $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{2n}$.

The hypothesis under test is

$$H_0 : S \text{ and } R \text{ are exchangeable}$$

given the nuisance features of §3.5, against the alternative that the model’s internal representation separates them beyond what the nuisance features permit. Rejecting H_0 is evidence that M represents S differently for reasons not reducible to how S and R differ as text.

3.2 The exchangeability requirement

Everything downstream depends on the construction of R , and no statistical machinery repairs a bad R . We state the requirement first because it is the binding constraint on the whole method.

Requirement E. R must be constructed so that, under the assumption of no contamination, an analyst told only the text of an item could not do better than chance at saying whether it came from S or R .

Three constructions satisfy Requirement E to different degrees.

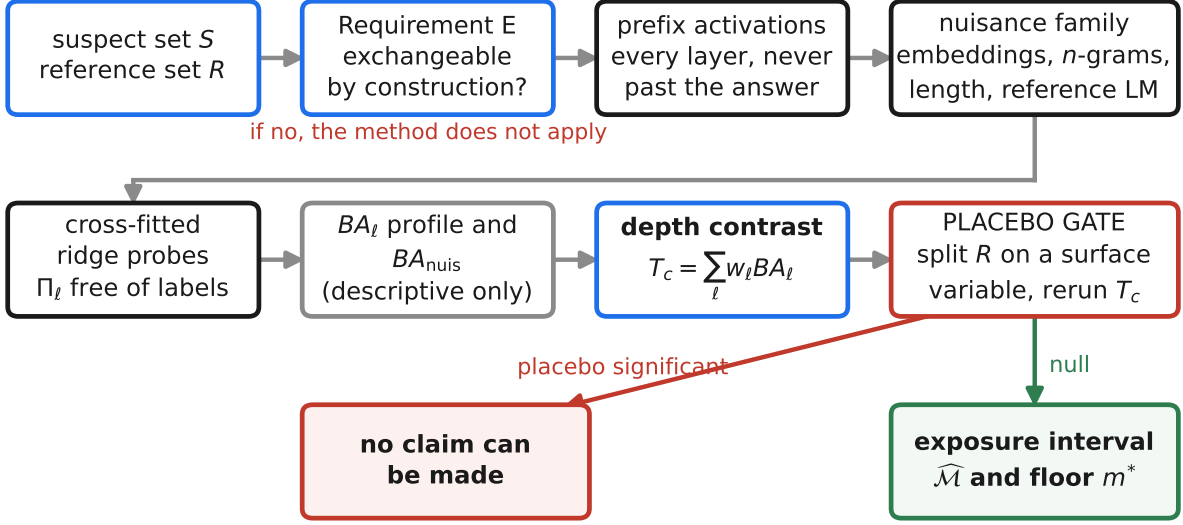


Figure 2: The protocol end to end. Two of the boxes can terminate the analysis rather than feed it: Requirement E, which most benchmarks fail for want of an exchangeable reference set, and the placebo gate of §4, which fails whenever surface separability grows with depth in the model under audit. Drawing the placebo as a gate rather than a control is the change §4 forced.

E1, randomised injection. Partition a single item pool at random into an injected half and a withheld half before training. Exchangeability holds by randomisation, exactly, and this is the only construction that is exact. It is available only when the analyst controls training, which is why the calibration stage of RSCP is built on it. The public checkpoints of Oren et al. [49] provide a pre-existing instance.

E2, commissioned twin. Have a fresh item set written to match the original on the covariates that drive surface separability. GSM1k [67] was commissioned this way against GSM8K [19], matched on human solve rate, number of solution steps, and answer magnitude. Exchangeability is approximate and rests on the matching being complete, which is checkable with the nuisance baseline itself.

E3, within-corpus held-out split. Use a corpus whose train and held-out partitions were drawn from the same distribution, as with Pile train versus Pile validation for the Pythia suite [6, 27]. Exchangeability is approximate and known to fail for some Pile subsets, since documents are ordered in time and some subdomains drift, which is one of the cases Das et al. [20] document. Per-subdomain analysis with the nuisance baseline reported alongside is mandatory here.

Two constructions that do *not* satisfy Requirement E, and which cover most existing membership-inference evaluation sets, are the temporal split, where R postdates the model cutoff and therefore differs in topic and vocabulary, and the cross-source split, where S and R come from different websites, licences, or extraction pipelines.

An important consequence is that RSCP cannot currently be applied to most benchmarks, because most benchmarks have no exchangeable twin. This is a property of the evaluation ecosystem rather than of the method, and it motivates the benchmark-design recommendation in §6.

3.3 Extraction points and the prefix rule

Define two pooling operators on a sequence x of length T :

$$E_{\text{last}}^{\ell}(x) = h_T^{(\ell)}(x), \quad (1)$$

$$E_{\text{mean}}^{\ell}(x) = \frac{1}{T} \sum_{t=1}^T h_t^{(\ell)}(x). \quad (2)$$

Applying each to the prefix p or the full item f gives four candidate extraction points. RSCP requires a prefix-based point, $\pi \in \{(\text{pre}, \text{last}), (\text{pre}, \text{mean})\}$, and forbids full-item points.

The argument is a conditioning argument. Let $Z \in \{0, 1\}$ be membership and $C \in \{0, 1\}$ the event that M answers the item correctly. Contamination inflates scores, so Z and C are dependent by construction; that dependence is the harm we are trying to quantify. The state $h_T^{(\ell)}(f)$ is computed from a sequence that contains a , and for ℓ close to L it is by definition a sufficient statistic for the model’s next-token distribution given $q \oplus a$. A probe on that state can therefore approximate a function of the model’s scoring of the gold answer, which is a proxy for C . Two failures follow.

The first is a confound. If S and R differ in difficulty at all, even slightly, a probe reading a proxy for C separates

them without any memorisation being present, and reports it as contamination.

The second is worse and survives perfect difficulty matching. Suppose S and R are matched on intrinsic difficulty but M is contaminated on S , so $\Pr(C \mid Z=1) > \Pr(C \mid Z=0)$. A full-item probe now separates the sets, but it does so by detecting the score inflation itself. The statistic is then a re-description of the accuracy gap, and cannot be used to explain it. The value of a representational detector is that it gives an independent line of evidence about *why* the accuracy gap exists, and that value is destroyed by any extraction point that can see the answer.

This is the same structure as the pre- versus post-continuation artefact identified in the author’s earlier layer-wise probing work on Japanese-English pragmatic reasoning [9], where probes read after a generated continuation absorbed information the model had itself produced, inflating apparent encoding of the target property. Here the feedback runs through the answer tokens rather than through a generated continuation, and the fix is the same: extract strictly upstream of any token whose identity or scoring depends on the property under test.

We recommend (pre, last) as the primary point, with (pre, mean) reported as a robustness check, and we recommend reporting the full-item points as well, clearly marked as diagnostics of the confound rather than as contamination estimates. The gap between the prefix and full-item statistics is itself informative: a large gap is a sign that whatever the full-item probe found is competence rather than familiarity.

3.4 The probe

Fix an extraction point π and a layer ℓ . Let $\phi_\ell(x) \in \mathbb{R}^d$ be the extracted activation, standardised feature-wise using training-fold statistics only. The probe is ridge regression on the ± 1 -coded label,

$$\hat{w}_\ell = \arg \min_{w \in \mathbb{R}^d} \|X_\ell w - \tilde{y}\|_2^2 + \lambda \|w\|_2^2, \quad (3)$$

where $X_\ell \in \mathbb{R}^{2n \times d}$ stacks the standardised activations and $\tilde{y} = 2y - 1$.

We use ridge rather than logistic regression for a reason that becomes important in §3.6: the fitted values are linear in the labels. Partition $\mathcal{D}$ into K folds and let $k(i)$ be the fold containing item i , let $X_\ell^{(-k)}$ stack the rows outside fold k , let P_{-k} be the selection matrix returning those rows to their positions in $\mathcal{D}$, and write

$$A_\ell^{(-k)} = (X_\ell^{(-k)\top} X_\ell^{(-k)} + \lambda I)^{-1}. \quad (4)$$

The cross-fitted smoother $\Pi_\ell \in \mathbb{R}^{2n \times 2n}$ has rows

$$(\Pi_\ell)_i = x_i^\top A_\ell^{(-k(i))} X_\ell^{(-k(i))\top} P_{-k(i)}. \quad (5)$$

By construction $(\Pi_\ell)_{ij} = 0$ whenever $k(i) = k(j)$, so every prediction is out of fold, and Π_ℓ does not depend on $\tilde{y}$. That second property is what §3.6 exploits.

Out-of-fold scores are $\hat{u}_\ell = \Pi_\ell \tilde{y}$. Writing $n_1 = \sum_i y_i$ and $n_0 = 2n - n_1$, the probe’s balanced accuracy is

$$\text{BA}_\ell(\tilde{y}) = \frac{1}{2n_1} \sum_{i:y_i=1} \mathbf{1}[\hat{u}_{\ell i} > \tau] + \frac{1}{2n_0} \sum_{i:y_i=0} \mathbf{1}[\hat{u}_{\ell i} \leq \tau] \quad (6)$$

with τ the median of $\hat{u}_\ell$, which keeps the statistic invariant to a constant shift and avoids a threshold chosen on the same data.

The regularisation λ is chosen on an inner split of the training folds only. Fold assignment is repeated over five seeds and all reported quantities are means with standard deviations over seeds, following the convention of the author’s earlier probing work.

3.5 The nuisance family and excess separability

Let Φ_0 be a declared family of model-independent or memorisation-independent feature maps:

- ψ_{bag} : mean-pooled layer-0 activations over the prefix. This is a bag of token embeddings with no attention and no MLP, so it carries no information the model computed. It is the natural floor for a representational claim.
- ψ_{char} : ℓ_2 -normalised character 3- to 5-gram TF-IDF.
- ψ_{surf} : token count, character count, digit ratio, punctuation ratio, type-token ratio, mean word length.
- ψ_{ref} : per-token log-likelihood summaries (mean, variance, minimum- $k\%$ mean for $k \in \{10, 20\}$, zlib-normalised ratio) under an independent reference model M_{ref} that has not been trained on any candidate corpus.

The nuisance baseline is the cross-fitted balanced accuracy of the same ridge probe on the concatenation of all four, using the same folds and the same seeds:

$$\text{BA}_{\text{nuis}} = \text{BA}(\text{concat}(\psi_{\text{bag}}, \psi_{\text{char}}, \psi_{\text{surf}}, \psi_{\text{ref}})). \quad (7)$$

Excess separability at layer ℓ is

$$\Delta_\ell = \text{BA}_\ell - \text{BA}_{\text{nuis}}. \quad (8)$$

Δ_ℓ is reported, but it is not tested. §4 measures the obvious statistic built on it, $\max_\ell \Delta_\ell$, and finds it unusable: BA_{nuis} depends on the dimension of the nuisance block, so the level of Δ_ℓ is biased in a direction set by an implementation detail. What is tested instead is a zero-sum contrast on the depth profile, recentred on a level-matched placebo baseline, given in Eq. (14).

Two points about ψ_{ref} . It is deliberately a strong adversary: it includes the statistics that likelihood-based membership inference uses [55], computed on a model that cannot have memorised the items. Whatever separability those statistics achieve on M_{ref} is attributable to how the

two sets differ as text. Setting the bar there is conservative, and it is the point of the construction. It also inherits a weakness: if M_{ref} is itself contaminated on S , BA_{nuis} is inflated and Δ_ℓ is biased downward, making the test conservative rather than anti-conservative. We prefer that direction of error and note it in §7.

Required controls. Three further checks accompany every reported Δ .

Control task. Replace y with labels drawn uniformly at random, independent of set membership, and rerun the whole pipeline. Selectivity is Δ_ℓ minus the control-task value [34]. A pipeline that produces positive selectivity on random labels has a leak, usually in the standardisation or fold construction.

Random-direction baseline. Replace ϕ_ℓ by its projection onto a random r -dimensional subspace for $r \in \{16, 64, 256\}$ and by projections onto single random directions. This checks that T is not an artefact of d being large relative to n , and follows the random-direction convention of the author’s earlier probing work.

Internal null. Split R at random into halves R_1, R_2 and run the entire protocol with $S \leftarrow R_1, R \leftarrow R_2$. This is an exact instance of E1, so T should be near zero and the permutation p -value should be uniform. Twenty replicates with a Kolmogorov-Smirnov test against uniformity is a cheap and effective end-to-end pipeline test.

Length matching. Stratify by prefix token length into deciles and subsample to equalise the two sets’ length distributions, then report the length-matched T .

3.6 Inference

Under H_0 the labels are exchangeable, which licenses a permutation test. Draw B permutations σ_b of $\{1, \dots, 2n\}$, form $\tilde{y}^{(b)} = \tilde{y}_{\sigma_b}$, and recompute

$$T^{(b)} = \max_\ell \left[\text{BA}_\ell(\tilde{y}^{(b)}) - \text{BA}_{\text{nuis}}(\tilde{y}^{(b)}) \right]. \quad (9)$$

The p -value is

$$\hat{p} = \frac{1 + \#\{b : T^{(b)} \geq T\}}{1 + B}. \quad (10)$$

The statistic actually tested is the recentred contrast of Eq. (14), a single pre-registered scalar, so no correction across layers is needed. What the permutation supplies that a bootstrap does not is the variance of the probe fit: Π_ℓ is label-free, so $\Pi_\ell \tilde{y}^{(b)}$ is the out-of-fold prediction of a probe refitted to the permuted labels, and §4 measures the difference this makes. When several benchmarks or models are audited, apply Benjamini-Hochberg across the resulting p -values [5].

Cost. Naively this requires refitting $B \times (L + 1) \times K$ probes. Equation (5) removes that. Because Π_ℓ is independent of the labels, it is computed once per layer per

fold-seed, at cost $O(K(nd^2 + d^3))$ or $O(Kn^3)$ via the dual form when $d > n$, and every permutation thereafter costs a single matrix-vector product $\Pi_\ell \tilde{y}^{(b)}$ at $O(n^2)$, plus a sort for the median threshold. For $n = 1000$ per set, $L = 24$, $K = 5$, $B = 1000$, the permutation stage is on the order of 10^{11} floating-point operations, which is seconds on a laptop. Activation extraction, done once, dominates.

This is the practical reason for choosing ridge over logistic regression. Logistic fitted values are not linear in the labels, so the shortcut does not apply and the permutation test becomes the expensive part of the protocol. In the linear-probe regime the two objectives generally rank layers similarly, but this is a design choice rather than an established fact and §10 lists it as something to check.

3.7 Exposure calibration

A binary contaminated/clean verdict is not the right output, because contamination is graded and because a null result at one exposure level says nothing about another. RSCP therefore reports an interval on effective exposure.

On a family of calibration models trained with known duplication counts $m \in \{0, 1, 2, 4, 8, 16, 32\}$ (construction E1), fit a monotone saturating link

$$g(m; \gamma, \beta) = \gamma(1 - e^{-\beta m}), \quad g(0) = 0, \quad (11)$$

to the observed (m, T) pairs, with γ the asymptotic excess separability and β the exposure sensitivity. The saturating form encodes the expectation that additional repetitions have diminishing effect once an item is memorised, consistent with the duplication-count dependence of memorisation reported by Carlini et al. [16] and Kandpal et al. [38].

For a suspect benchmark with observed T and permutation-derived null spread, the reported quantity is the set

$$\widehat{\mathcal{M}} = \{m : T \in \text{CI}_{95}(g(m; \hat{\gamma}, \hat{\beta}))\}, \quad (12)$$

read as “the observed excess separability is consistent with an effective exposure of $\widehat{\mathcal{M}}$ repetitions”. Two quantities from the calibration are worth reporting on their own. The *detection floor* m^* is the smallest m at which the test rejects at $\alpha = 0.05$ with power 0.8 at the given n ; it states what the instrument can and cannot see. The *saturation point* is the m beyond which g is flat, beyond which the method cannot distinguish heavier contamination.

Transfer of $\hat{\gamma}, \hat{\beta}$ from calibration models to a target model is an assumption, not a result, and is the weakest link in the procedure. We recommend calibrating within model family and scale wherever possible, and reporting $\widehat{\mathcal{M}}$ only when a same-family calibration exists.

3.8 Item margins and contamination-adjusted scores

The out-of-fold score $\hat{u}_{\ell^*i}$ gives a per-item margin. Individually these are noisy, for the reasons Duan et al. [24] doc-

1	Construct R satisfying Requirement E, with $ R = 2 S $.
2	Extract prefix activations at all layers, (pre, last) primary.
3	Fit the nuisance probe; report BA_{nuis} before anything else.
4	Fit cross-fitted ridge probes at each layer over 5 fold-seeds.
5	Report Δ_ℓ and the profile BA_ℓ descriptively.
6	Build the level-matched placebo baseline on the other half of R .
7	Test the recentred contrast (Eq. 14) against the permutation null.
8	Run the four controls: control task, random direction, internal null, length matching.
9	If a same-family calibration exists, report $\widehat{\mathcal{M}}$ and m^* .
10	Report the full-item extraction points as confound diagnostics.
11	If reporting adjusted scores, report both raw and difficulty-matched inflation, with the panel.

Table 1: The RSCP protocol. Steps 1, 3, 6 and 7 are the ones §4 shows cannot be skipped.

ument, and we do not propose per-item verdicts. Ranked, they support a set-level comparison.

Let S_{hi} and S_{lo} be the top and bottom margin quartiles of S . The raw inflation estimate is

$$\widehat{\text{Infl}} = \text{acc}(S_{\text{hi}}) - \text{acc}(S_{\text{lo}}). \quad (13)$$

This is circular if margin correlates with item difficulty, which it plausibly does, since easy items are shorter, more templated, and more likely to recur online. The control is difficulty matching against a panel. Estimate a difficulty score b_i for each item as the fraction of a panel of reference models, chosen to predate the benchmark or to have verified clean corpora, that answer it correctly, then resample S_{hi} and S_{lo} to equalise the distribution of b and recompute. Report both, and report the panel. The difficulty estimate can equivalently be an item-response-theory difficulty parameter fitted across the panel, which is the natural point of contact between this method and psychometric approaches to benchmark analysis [51].

Regression to the mean will produce a positive raw $\widehat{\text{Infl}}$ even with no contamination whenever accuracy is measured on the same items used to fit the margin, so the accuracy evaluation must be done on a held-out fold, using the same fold structure as the probe.

3.9 The protocol as a checklist

Table 1 states the procedure compactly. Steps 1, 3 and 7 are the ones most often skipped, and a result reported without them should be read as an upper bound on the memorisation signal rather than an estimate of it.

	Sim-A	Sim-B
residual-stream norm growth	$\times 1.00$	$\times 1.75$
correlation, layer 1 to L	0.01	0.65
kurtosis of activations	3.0	9.3
share of variance, top eigenvector	0.025	0.188

Table 2: The two simulators occupy different regimes. Sim-B adds anisotropic power-law covariance, heavy-tailed innovations, growing per-layer scale, several correlated nuisance directions, and accumulation down the residual stream.

4 Validation, and the Four Choices It Forced

Everything above the line is a proposal. One part of it can be settled without a GPU: whether the statistical procedure does what it claims. This section reports that, and it is the reason the protocol looks the way it does. Each of the four design choices in §3 was adopted because a simpler alternative was measured and failed.

Measurements of a statistical procedure are only as good as the range of data they were taken over, so everything is run under two generative models. Sim-A is deliberately simple: isotropic Gaussian layers, one nuisance direction, layers conditionally independent. Sim-B is a residual stream with the pathologies real transformers have. Table 2 shows they are in different regimes, which is a measurement rather than an assertion. Where a conclusion holds under both we report it as a property of the procedure; where it holds under one only, we say so.

In both, the nuisance is a continuous item covariate whose distribution differs between the two sets, not a property of membership. That detail is load-bearing. If surface variation were tied to the label, depth-dependent surface separability would exist only along the suspect-reference axis, no within-reference split could ever observe it, and a placebo baseline would be useless by construction. An earlier version of these simulators made exactly that mistake and produced a misleading negative result.

Throughout: $d = 96$, $L = 12$, 200 suspect items, twice that many reference items, 300 permutations, $\alpha = 0.05$, and 40 to 60 replicates per cell. The whole study is about 8 minutes on four CPU cores.

4.1 C1: the level of separability is unusable

The obvious statistic is excess separability, $\Delta_\ell = \text{BA}_\ell - \text{BA}_{\text{nuis}}$, aggregated over layers. It does not work, and the reason has nothing to do with contamination. At matched effective degrees of freedom a lower-dimensional feature block still estimates its discriminative direction from a smaller space and reaches higher accuracy at equal population separation, so BA_{nuis} depends on how large the analyst made the control set, and Δ_ℓ inherits that as a bias.

Table 3 and Figure 3 measure it. Under a true null the max-level statistic rejects 0.030 of the time against a

Statistic	control set dimension p		
	96	400	1200
raw accuracy, no control	1.000	1.000	1.000
$\max_{\ell} \Delta_{\ell}$ (level)	0.030	0.530	0.988
$\overline{\Delta}_{\ell}$ (level)	0.030	0.640	0.975
depth contrast	0.020	0.000	0.025
BA_{nuis}	0.652	0.609	0.570

Table 3: **C1.** False positive rate under a true null as the nuisance control set grows, Sim-A, nominal 0.05. The level of excess separability inherits the control set’s estimation capacity; the zero-sum contrast cancels it algebraically.

Simulator	depth slope	uncorrected	recentred
Sim-A	0%	0.017	0.083
	25%	0.283	0.050
	50%	0.717	0.050
Sim-B	0%	0.000	0.083
	25%	0.000	0.083
	50%	0.000	0.033

Table 4: **C2.** False positive rate under a true null when surface decodability grows with depth. Recentring on the level-matched placebo holds the nominal rate where the uncorrected contrast does not.

dimension-matched control and 0.988 against a control of $p = 1200$. The direction is what matters: the nuisance family we recommend contains a character n -gram block far larger than the activations, so a protocol reporting the level would sit at the right-hand end of that figure and manufacture contamination on clean models. Subtracting BA_{nuis} does not repair the naive statistic, which rejects 1.000 everywhere; it converts a guaranteed error into one that depends on an arbitrary implementation choice, which is worse because it looks principled.

A contrast with weights summing to zero removes the problem exactly, since $\sum_{\ell} w_{\ell}(BA_{\ell} - BA_{\text{nuis}}) = \sum_{\ell} w_{\ell}BA_{\ell}$. It also changes what is being tested, from the level of separability to its shape in depth, which is the quantity that distinguishes memorisation from surface difference in the first place.

4.2 C2: depth profiles are not flat, and fail both ways

A contrast against a fixed rising ramp assumes surface decodability is flat in depth. That assumption fails, and it fails in both directions.

Make surface separability grow with depth and the uncorrected contrast rejects a true null (0,0.01667) (0.25,0.2833) (0.5,0.7167) of the time at 25 per cent growth and 0.717 at 50 per cent. Run instead on a residual stream where the surface signal is written once at the embedding and noise accumulates on top of it, and the profile declines:

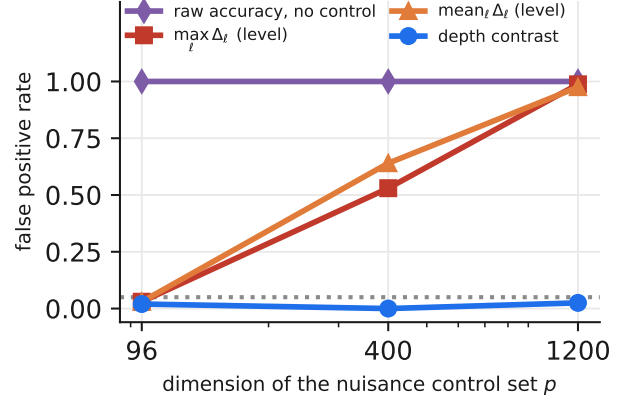


Figure 3: **C1.** False positive rate under a true null as the nuisance control set grows. The zero-sum contrast is flat because the control term cancels algebraically; every level-based statistic tracks the control set’s dimension.

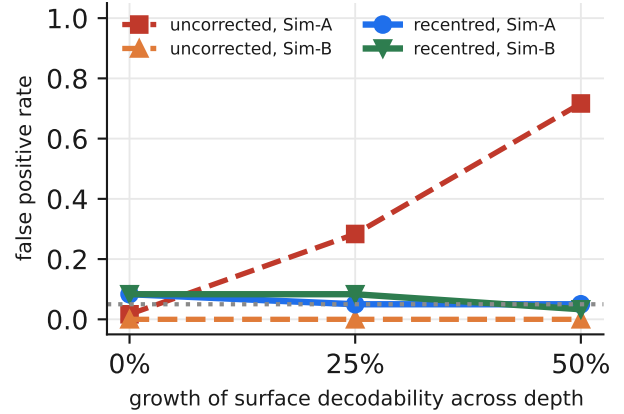


Figure 4: **C2.** Recentring holds the nominal rate across the whole range where the uncorrected contrast fails. Measured under both simulators, true null throughout.

under Sim-B a true null falls from 0.754 at the embedding to 0.657 at the final layer (Figure 6). A declining baseline makes the contrast negative before any memorisation is present, and the test loses all of its power rather than gaining false positives.

The fix does not add an assumption, it removes one. The placebo split already estimates the baseline: two halves of the reference set, every item a non-member, so whatever depth profile appears there is the profile of surface separability in this model. Recentring on it,

$$T_{\text{adj}} = \sum_{\ell} w_{\ell}(BA_{\ell} - BA_{\ell}^{\text{plc}}), \quad (14)$$

replaces an untestable global claim with a locally estimated correction. The placebo stops being a veto and becomes the baseline.

Table 4 and Figure 4 give the result. Recentring holds the false positive rate at 0.050 where the uncorrected statis-

tic reaches 0.717, and it does so without costing power: Sim-B, where the uncorrected contrast had none, reaches 0.780 at $\varepsilon = 1.5$.

4.3 C3: the null must refit the probe

The natural null for a statistic computed on items is a bootstrap over items. It is wrong here. Resampling items with the fitted probe held fixed captures evaluation variance but not the variance of the fit, and the omission is not small: under a true null the item bootstrap rejects between 0.030 and 0.087 of the time across the same cells where the permutation null holds 0.020 to 0.025.

Permuting labels against the cross-fitted smoother captures both, because Π_ℓ is label-free and $\Pi_\ell \tilde{y}^{(b)}$ is therefore the out-of-fold prediction of a probe *refitted* to the permuted labels. The same property that makes the permutation cheap is what makes it correct. The measured rate is 0.020 to 0.025 across the range in Figure 3.

4.4 C4: the baseline needs the same sample size

A baseline estimated on half as many items as the comparison is not comparable, because probe accuracy depends on sample size and the dependence differs across the profile. Splitting the reference set in half to build the placebo, while comparing at full size, gives a false positive rate of 0.217 under Sim-A against 0.067 when the sizes match. The protocol therefore requires $|R| = 2|S|$: half the reference set is the comparison arm, half is the baseline arm. Under Sim-B the two agreed, so this binds in some regimes and not others, and we impose it because it is free at design time.

Matching size is necessary and not sufficient. Splitting the reference set at the extremes of a surface key separates the halves more strongly than the two sets actually differ, and a stronger separation bends the profile differently. We therefore search over how coarsely to split and take the split whose embedding-layer separability is closest to the observed value. Layer 0 is the right anchor because it is a bag of token embeddings and carries surface information only.

4.5 What the instrument can see

Figure 5 gives power against planted effect size. Both simulators cross 0.8 near $\varepsilon = 1.5$ (0.860 and 0.780) and reach 1.000 by $\varepsilon = 3$. Below $\varepsilon \approx 1$ the instrument is underpowered, which is the resolution it has and the reason a null result has to be reported with a detection floor beside it.

Figure 6 shows the profiles the contrast weights, and makes the C2 problem visible directly: the two panels have null profiles of opposite slope, and only a baseline estimated per model can accommodate both.

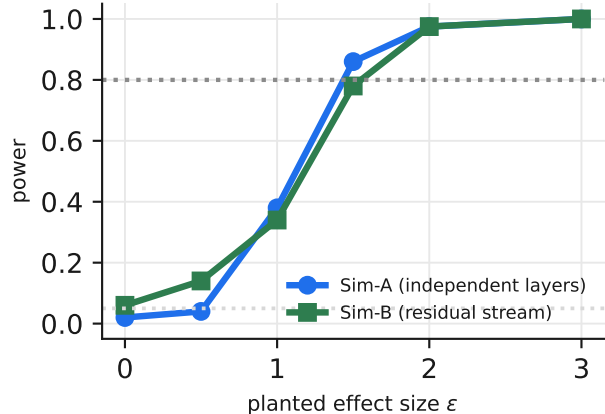


Figure 5: Power of the recentred contrast under both simulators. Both reach 0.8 near $\varepsilon = 1.5$ and saturate by $\varepsilon = 3$.

4.6 The placebo is only as good as the surface key

The baseline is built by splitting the reference set on a variable the analyst chooses, and that choice is not neutral. Degrading the key from a close correlate of the underlying covariate to a loose one takes power from 0.825 to 0.450 while the false positive rate stays at or below 0.050 (Figure 7). The failure mode is therefore benign: a poor key makes the instrument blind, not wrong. This is worth stating plainly because the opposite would make the method unusable in practice, where the analyst rarely knows which surface feature separates the two sets.

4.7 Cost

At $n = 1000$ items per set and $d = 2048$, building the cross-fitted smoother takes 6.51 s, after which each permutation costs 1.19 ms and 2000 of them cost 2.4 s. Refitting per permutation, which any probe without label-linear fitted values forces, projects to 1.9 hours, a factor of about 2919. That is the difference between running the control battery and skipping it, and it is why the battery is mandatory here rather than encouraged.

4.8 Honest residuals

Three things are not fixed. Calibration is close to nominal but not exact, sitting between 0.083 and 0.083 across the null cells, so p -values near the threshold should not be read as exact. Power under a realistic residual stream requires a larger effect than under independent layers, and we have no way to convert either ε scale into a duplication count without the injection experiments of §10. And every number here is a measurement of a procedure on synthetic activations. None of it is evidence that a real transformer carries a familiarity direction at all.

5 Worked Examples

Three settings are described. The first calibrates the instrument against a corpus whose contents are known. The

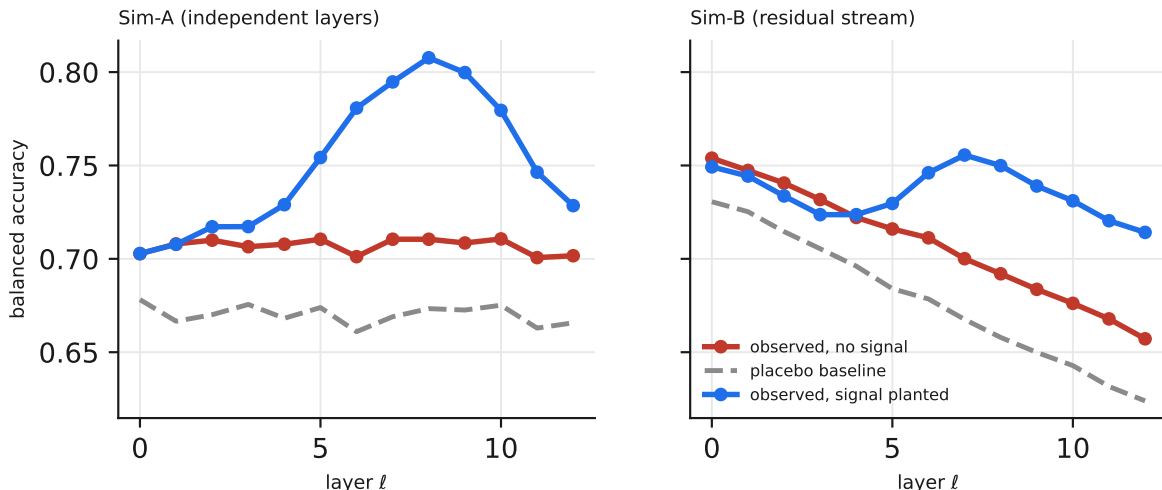


Figure 6: Recovered depth profiles. Left, independent layers: the null profile is flat and the placebo tracks it. Right, a residual stream: the null profile declines by roughly nine accuracy points from embedding to final layer, which is the failure mode C2 corrects. In both, the planted signal is recovered with its peak near the planted layer.

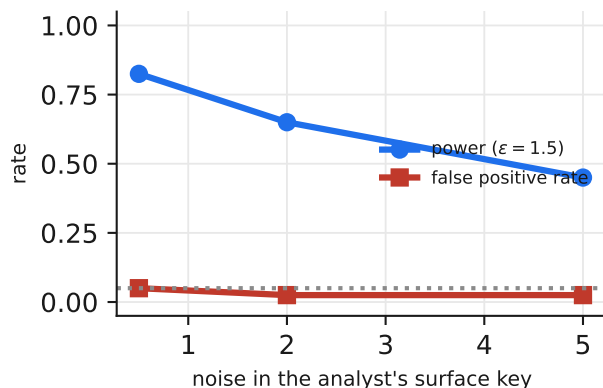


Figure 7: Degrading the analyst’s surface key costs power and does not cost validity. Measured under Sim-B.

second uses existing public checkpoints with deliberately injected contamination at known duplication counts. The third applies the protocol to a benchmark of live interest.

Numbers attributed to published work are real measurements and are cited. Numbers in tables marked *illustrative* are synthetic. They are there to show the shape of the output and to make the reporting format concrete; the study in §10 would produce the real values.

5.1 WE1: Pythia and the Pile

Why this setting. The Pythia suite [6] trains sixteen models from 70M to 12B parameters on the Pile [27] in a fixed data order, releases 154 checkpoints per model, and releases tooling to reconstruct the exact training dataloader. Three properties follow that no other public suite offers together. Membership is known exactly. The step at which a sequence was seen is known, which allows expo-

sure recency to be varied. And the deduplicated variants of each model give a second, independent handle on duplication count, since deduplication removes exactly the repeated exposures that drive memorisation [39]. Which sequences a Pythia model memorises is partly predictable from smaller models in the same suite [7], which gives an additional external quantity for T to be validated against. Seed-level variability in this suite has been characterised directly [61], so the fold-seed variance we report can be compared against the pretraining-seed variance that bounds it.

Design. Take S from the Pile training split and R from the Pile validation split, matched within subdomain, so construction E3 applies. Run per subdomain across the drift spectrum, PubMed Central and Wikipedia at the stable end, GitHub and arXiv at the drifting end, and report BA_{nuis} per subdomain first. Extract at (pre, last) for Pythia-410M, 1.4B and 2.8B and their deduplicated counterparts at every layer. Storage is modest: 2000 items times 25 layers times 2048 dimensions in half precision is about 200 MB per model.

Expected outcome. Duan et al. [24] evaluate this exact setting for likelihood-based membership inference across the Pythia suite and find performance barely above chance, which they attribute to the very large corpus, close to a single pass over it, and a fuzzy member boundary. The honest prior for WE1 is therefore that T is small. That is not a failure of the experiment. WE1’s purpose is to establish the noise floor and to confirm that the internal-null check produces uniform p -values on a real corpus, which is the

Method	PubMedQA ($m=1$)	CommonsenseQA ($m=2$)
Loss attack	48.0	49.9
Neighbour attack	53.0	50.0
Min- k % Prob	47.5	49.6
zlib ratio	46.1	48.8
Lowercase ratio	50.7	49.2
Smaller-model ratio	49.5	49.5
Activation probe	54.0	51.9

Table 5: Published AUC (%) on the deliberately contaminated checkpoints of Oren et al. [49], as measured by Liu et al. [42], Table 4. Chance is 50.0. These are real measurements, not illustrative. m is the injection duplication count.

precondition for believing anything the protocol says in WE2 and WE3.

Recency sub-experiment. Because the training step at which each sequence was seen is known, S can be stratified by exposure step and T computed per stratum, giving a forgetting curve for representational familiarity. This connects to the causal treatment of memorisation profiles over Pythia’s data order given by Lesci et al. [40] and to the forgetting dynamics of Jagielski et al. [35]. A monotone decay of T with time since exposure would be strong evidence that T is tracking memorisation rather than a static property of the text, and is the single most convincing validity check available in this setting.

5.2 WE2: deliberately contaminated checkpoints

Why this setting. Oren et al. [49] released models pre-trained with intentional injection of benchmark data, at 1.4B and 124M parameters, together with the injected datasets and their duplication counts. Because injected and clean items come from the same benchmarks and the assignment is not temporal, this is an instance of E1 and exchangeability holds without further work. As reported by Liu et al. [42], PubMedQA [36] was injected once and CommonsenseQA [59] twice.

Published reference point. Table 5 reproduces the AUCs Liu et al. [42] measured on these checkpoints. Every method, including their activation probe, sits within a few points of chance. This is the most informative published result on what activation probing can do under an exchangeable split at realistic duplication counts, and it should anchor expectations at $m \in \{1, 2\}$.

Design. Apply RSCP with S the injected items and R the withheld items of the same benchmark. Report Δ_ℓ across all layers of the 1.4B model, with the 124M model as a scale contrast. Two extensions matter.

m	Verbatim	Eval-fmt	Paraphrase	Source doc
0	0.00	0.00	0.00	0.00
1	0.01	0.02	0.00	0.00
2	0.02	0.03	0.01	0.01
4	0.05	0.07	0.02	0.01
8	0.11	0.14	0.05	0.02
16	0.18	0.22	0.09	0.04
32	0.23	0.27	0.13	0.06

Table 6: *Illustrative.* Shape of the exposure-response table for WE2 (T , Pythia-410M continued pretraining). Values are synthetic and are shown only to define the reporting format and to make the fitted link in Eq. (11) concrete. The quantities of scientific interest are the detection floor m^* in each column and the ordering of the four columns, both of which the study in §10 would determine.

First, extend the exposure range. The public checkpoints give only $m \in \{1, 2\}$, which is exactly the regime where everything is near chance and where the calibration curve is least identifiable. Continued pretraining of Pythia-160M and Pythia-410M on a held-out slice of the Pile with a benchmark injected at $m \in \{0, 1, 2, 4, 8, 16, 32\}$ fills in the curve. Randomised assignment of items to injected and withheld halves gives exact exchangeability.

Second, vary the injection format, since contamination in the wild does not arrive as clean question-answer pairs. Four conditions: verbatim item text; item text in evaluation format with the gold answer; a paraphrase from an independent model, following Yang et al. [64]; and the source document without question framing. The paraphrase condition is what distinguishes RSCP from overlap search, which is blind to it by construction.

The deliverable of WE2 is the detection floor m^* per condition, the number that tells practitioners when an activation probe is the wrong tool.

5.3 WE3: GSM8K against GSM1k

Why this setting. GSM1k [67] is a set of 1000 grade-school arithmetic problems commissioned to mirror GSM8K [19], matched on human solve rate, number of solution steps, and answer magnitude. It is the clearest existing instance of construction E2. Its authors report accuracy drops of up to 8 points on GSM1k relative to GSM8K, with several model families showing systematic gaps across sizes, and a positive relationship (Spearman $r^2 = 0.36$) between a model’s probability of generating a GSM8K example and its GSM8K-to-GSM1k gap. Frontier families show little or no gap.

What RSCP adds. The GSM1k evidence is behavioural. A gap on a matched fresh set is consistent with memorisation, but equally with imperfect matching, with GSM1k being harder along an unmatched dimension, or with over-

fitting to GSM8K’s surface conventions. These have different consequences for how to read a GSM8K number, and behavioural evidence cannot separate them.

RSCP gives a second, representational line of evidence on the same split, and the two lines can disagree in an informative way:

Figure 8 sets out the four outcomes and what each licenses.

Model selection. Three open models runnable on one consumer GPU: an OLMo model, because Dolma is public and any verdict can be checked against corpus search [26, 30, 57]; a Pythia model of comparable scale as a low-contamination control, since the Pile predates GSM8K’s wide circulation; and one recent small instruction-tuned model, where the prefix rule applies inside the chat template and extraction is at the last token before the assistant turn.

A check that costs nothing. Before any probing, compute BA_{nuis} for GSM8K against GSM1k. If it is near 0.5 the commissioned matching worked, which is an independent audit of GSM1k that its own construction checks do not provide. If it is well above 0.5, a blind classifier can tell the two sets apart, every behavioural comparison between them carries a caveat, and the responsible features can be read off the fitted nuisance weights. Either result is worth reporting.

6 Discussion

6.1 What the protocol says about the benchmarks it is applied to

On GSM8K. The field currently treats the GSM8K-to-GSM1k gap as the strongest available evidence that GSM8K is contaminated for some model families. The gap is real and carefully measured. What it does not do is distinguish memorisation of items from adaptation to a format. RSCP makes the distinction operational, and the negative branch is as useful as the positive one: a large gap with T near zero, on a model where the calibration curve says m^* is small, would mean the gap is not item memorisation, which would redirect attention from data hygiene to the more awkward question of whether heavy tuning on GSM8K-style data is itself a form of benchmark-specific overfitting that no amount of decontamination fixes.

On MMLU-style benchmarks. MMLU [32] was assembled from many independent online sources across 57 subjects, so its items have heterogeneous provenance and there is no reason to expect uniform contamination. The natural unit of analysis is the subject, not the benchmark, and RSCP is applied per subject with Benjamini-Hochberg

across subjects. The absence of an exchangeable reference set is the blocker: a fresh matched set would have to be commissioned per subject. Until then, MMLU can be audited only in the E1 direction, by injecting a randomly chosen half into a small model and measuring what exposure level would be needed to produce the separability seen in the wild.

On the Pile-derived settings. The most likely outcome of WE1 is a set of small, mostly non-significant T values with large BA_{nuis} on the drifting subdomains. Reported as excess separability rather than as raw accuracy, this is a clean replication of Duan et al. [24] through a different instrument, and it puts a number on how much of the apparent separability in this literature is corpus drift.

6.2 The most informative negative result

If the calibration curve of §3.7 places m^* above the duplication counts that occur in practice, the correct conclusion is that activation probing does not solve low-duplication benchmark contamination, and the paper reporting that should say so plainly. The published evidence points that way: likelihood-based attacks under exchangeable splits perform near chance [24], and the one published activation probe evaluated on a non-temporal split reached 54.0 and 51.9 AUC at $m = 1$ and $m = 2$ [42]. A well-instrumented negative result with a stated detection floor is more useful to the field than another detector reporting high AUC on a split that a blind classifier can also solve.

6.3 Implications for benchmark design

Three changes would make contamination auditing tractable, and all are cheap at construction time relative to their value later.

Publish a matched twin at release. Reserve a randomly chosen fraction of items at construction and release them separately, or on request under an agreement not to redistribute. This converts every future audit from construction E2, which requires commissioning new items and defending the matching, to construction E1, where exchangeability holds by randomisation. GSM1k [67] shows the value of the twin and also shows the cost of producing one after the fact.

Publish a canary. BIG-bench [58] includes a canary string for the purpose and the practice has not spread. None of ARC [18], HellaSwag [66], MATH [33] or MMLU carries one, and all four feed aggregate leaderboards [8], so contamination findings on them have to be argued indirectly.

Publish the matching statistics. With construction documented in enough detail, a BA_{nuis} audit can be run by anyone against any candidate reference set.

	accuracy gap on the matched twin	
	absent	present
depth contrast significant	Seen, not exploited Items are stored but not used. Memorisation without exploitation.	Contamination confirmed The straightforward reading. Report the exposure interval and the adjusted score.
depth contrast null	No evidence At this sensitivity. Report the detection floor so the null reads at the right strength.	Inflation, not memory Exposure below the detection floor, or format overfitting, or difficulty mismatch.

Figure 8: The four outcomes and what each licenses. Behavioural evidence alone cannot separate the right-hand column, which is what the representational test adds.

6.4 Relation to sample-efficient evaluation

Item margins from §3.8 produce a per-item score that is not a difficulty estimate and not a discrimination estimate, so it is orthogonal to the psychometric quantities used to prune benchmarks to an informative core [51]. A pruned benchmark selected purely for information content could be systematically more or less contaminated than the full set, since the item properties that make an item informative are correlated with the properties that make it likely to have been scraped. Auditing pruned benchmarks for this is a small, tractable study and we list it in §10.

7 Limitations and Threats to Validity

L1. Applicability is bounded by the reference set. RSCP cannot be run on a benchmark with no exchangeable twin, which today is most of them. Every refinement in §3 is downstream of a precondition the evaluation ecosystem mostly does not satisfy.

L2. White-box only. Activations are required, so closed models are out of scope; the black-box test of Oren et al. [49] remains the right tool there, with its own restriction to file-level contamination.

L3. A null result is weak evidence of absence. A linear probe lower-bounds decodability at a chosen extraction point [4, 50]. $T \approx 0$ means the familiarity signal is not linearly decodable from prefix activations, which is compatible with it being present nonlinearly, at a different extraction point, or in a subspace the standardisation sup-

presses. The detection floor m^* should always be reported alongside a null so the null can be read at the right strength.

L3b. The baseline is only as good as the surface key.

The placebo is built by splitting the reference set on a variable the analyst chooses, and a key that does not track whatever actually differs between the two sets gives a baseline describing the wrong null. §4 measures the direction of that failure and it is benign, costing power (0.825 down to 0.450 as the key degrades) rather than validity. But a weak key can make the instrument blind without announcing it.

L3c. Calibration is close but not exact. Across the null cells the recentred contrast rejected between 0.083 and 0.083 of the time rather than exactly 0.05. Reported p -values near the threshold should not be read as exact, and a headline claim should use a stricter level.

L3d. The simulators are not transformers. Sim-B has heavy tails, anisotropy, norm growth and residual accumulation, and it is still a generative model we wrote. Real activations may fail in ways neither simulator represents, and only Phase 1 will show that.

L4. The nuisance family is finite. $\Delta_\ell > 0$ says the probe beat the nuisance features we declared. It does not say the probe beat every model-independent feature of the text. The statistic is evidence, not proof, and the declared family should be published so others can attack it by adding

features. We consider this preferable to the alternative of an undeclared implicit baseline of chance.

L5. Calibration transfer is assumed, not shown. Converting T to $\widehat{\mathcal{M}}$ requires that $(\hat{\gamma}, \hat{\beta})$ estimated on a small calibration model apply to the target. Memorisation grows with scale [16, 60], so a calibration fitted at 410M will not transfer to a 70B model, and probably not cleanly across architectures at the same scale. Report $\widehat{\mathcal{M}}$ only with a same-family calibration; otherwise report T and $\hat{p}$ alone.

L6. Post-training breaks the clean story. Instruction tuning shifts representations and contamination can enter at that stage. The prefix rule also interacts with chat templates, where template tokens contribute a large constant component to the activation. Calibration models built with pretraining-stage injection only will understate post-training contamination.

L7. Item-level margins are unreliable. Per-item membership inference on pretraining data is close to undetectable [24]. The adjusted-score procedure in §3.8 uses margins only in aggregate over quartiles, and should not be reinterpreted as a per-item verdict, however tempting the ranking looks.

L8. Circularity in the adjusted score. Margin plausibly correlates with difficulty. The difficulty-matched control depends on the reference panel being uncontaminated, which is the assumption under audit, applied one level up. The matched estimate is better than the raw one and is not free of the problem.

L9. Adversarial evasion is easy. A developer wanting to hide contamination can remove linear decodability without removing the benefit, since nullspace projection and related methods delete a linearly encoded property while leaving behaviour largely intact [25, 52]. RSCP is an instrument for analysing models one is not adversarial towards. It is not an audit that survives a motivated adversary and should not be presented as one.

L10. Small models may not show the phenomenon. If the familiarity direction emerges only above some scale, results at 160M to 2.8B will be uninformative about the models people actually evaluate. The scale trend across the Pythia suite is the diagnostic, and a flat or non-monotone trend should be reported as failure to establish the effect rather than smoothed over.

L11. Multiplicity. The contrast is a single pre-registered test per model and item pool, but auditing many bench-

marks, models and extraction points reintroduces multiplicity; fix the analysis plan in advance and apply Benjamini-Hochberg across the pre-registered family [5].

L12. Statistical power. With n in the low thousands and effect sizes of a few balanced-accuracy points, power is limited (§4), and underpowered designs produce unreliable literatures [12, 23]. Power should be computed from the calibration curve before collection; a study that cannot reach 0.8 at the target m should say so rather than run.

L13. Dual use. A reliable membership detector applied to individuals’ text is a privacy attack [14, 48]. The set-level framing limits this, but item margins are individual scores and the calibration would transfer. Released artefacts should be restricted to benchmark items and public-corpus documents.

8 Phase 1: the Protocol on Real Models

Everything to this point is either a proposal or a measurement of a procedure on synthetic data. This section reports the protocol run against real transformers: 6 audits over 2 Pythia scales and two sources, about an hour of compute on a laptop. It is a small study and we draw correspondingly narrow conclusions, but one of them settles a question §4 could not.

8.1 Surface separability of published splits

The cheapest thing the protocol produces is BA_{nuis} , the accuracy of a classifier that never queries the model. On WikiMIA it is 0.855 to 0.856. The two sets are a temporal split, so they differ in topic and vocabulary, and a blind classifier separates them almost perfectly. This reproduces the finding of Das et al. [20] through a different instrument and makes its consequence concrete: no statistic computed on that split, ours included, can be read as evidence about memorisation. The protocol refuses the verdict rather than reporting it (2 of 6 audits).

Pile train against Pile validation gives 0.490 to 0.521, which is chance. For the two subdomains we looked at, at 500 items per set, construction E3 delivers what it promises. That is worth recording because Das et al. [20] document drift in some Pile subdomains, and it is not universal.

8.2 Real depth profiles are not flat, and the correction scales

§4 showed on synthetic data that a contrast against a flat depth profile fails in both directions, and could not say which way real transformers go. Figure 9 answers it. The profiles are not flat. On the temporal split they span 12.6 to 29.1 accuracy points and are hump-shaped, rising steeply to a mid-network peak and falling away. A test built on the assumption that surface decodability is constant in depth

Model	Split	n_S	BA_{nuis}	baseline span	T_{raw}	baseline	T_{adj}	Outcome
pythia-160m	pile github	500	0.490	3.9 pts	-0.0099	+0.0116	-0.0215	null, $p = 1.00$
pythia-410m	pile github	500	0.494	3.1 pts	-0.0067	+0.0048	-0.0114	null, $p = 0.96$
pythia-160m	pile wikipedia	500	0.517	2.9 pts	-0.0040	+0.0066	-0.0106	null, $p = 0.94$
pythia-410m	pile wikipedia	500	0.521	2.7 pts	-0.0048	+0.0050	-0.0098	null, $p = 0.94$
pythia-160m	wikimia	394	0.856	29.1 pts	+0.0498	+0.0138	+0.0361	no verdict, Req. E fails
pythia-410m	wikimia	394	0.855	12.6 pts	+0.0610	+0.0153	+0.0457	no verdict, Req. E fails

Table 7: Phase 1, measured. BA_{nuis} is what a classifier with no access to the model achieves; near 0.5 the two sets are exchangeable, near 0.86 they are not and no contamination verdict is admissible. The baseline span is the range of the placebo depth profile, and it is the quantity the flat-profile assumption gets wrong. Every Pile arm is null with a negative adjusted statistic, which is what the literature predicts for a corpus seen approximately once.

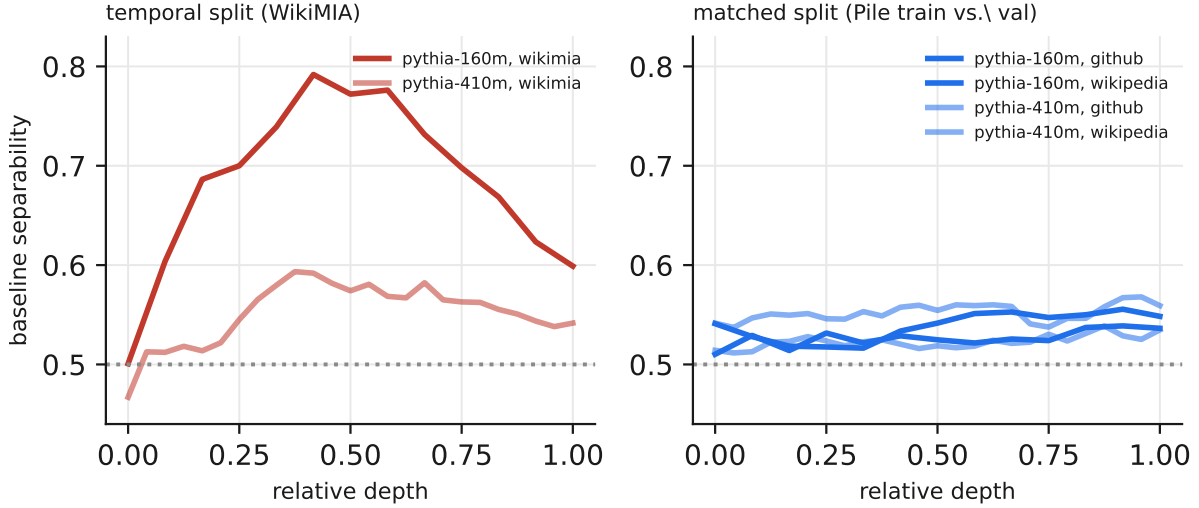


Figure 9: Baseline depth profiles measured on real models. Each curve is the separability of two halves of a reference set, where every item is a non-member, so it is the profile of *surface* separability in that model. The flat-profile assumption that §4 had to discard is the claim that these are horizontal lines.

would be comparing against a line that is wrong by up to 29.1 points.

The more useful finding is that the non-flatness tracks the surface difference. Across the 6 audits the correlation between BA_{nuis} and the baseline span is 0.87 (Figure 10). Where the two sets are well matched the baseline is nearly flat, 2.7 to 3.9 points, and the correction is close to a no-op. Where they are not, the baseline is large and absorbs 25 to 28 per cent of the raw statistic. A correction that is largest exactly where it is most needed, and negligible where it is not, is the behaviour one wants; the alternative would be a knob that moves results around indiscriminately.

8.3 The negative control behaves

All 4 Pile arms return null, at both scales and on both sub-domains, with a negative adjusted statistic in every case. This is what the literature predicts: Pythia sees the Pile approximately once, and Duan et al. [24] found membership inference near chance in exactly this setting. An instrument

that manufactured positives on well-matched sets would have shown it here, and did not.

The null has to be read at the right strength. Without the exposure calibration of §10 we cannot convert it into a statement about duplication counts, so the correct reading is that no depth-dependent familiarity signal is detectable at this sensitivity, not that none exists.

8.4 A split where Requirement E holds by construction

Every arm above rests on an exchangeability argument. Oren et al. [49] provide one that does not. They injected a random subset of PIQA’s test file into the pretraining corpus of a 1.4B model at a known duplication count and released the whole file alongside the checkpoint, so the withheld items are recoverable by difference. Both arms are then one pool split at random before training, which is construction E1: exchangeability holds by randomisation rather than by argument. All 500 injected items used here

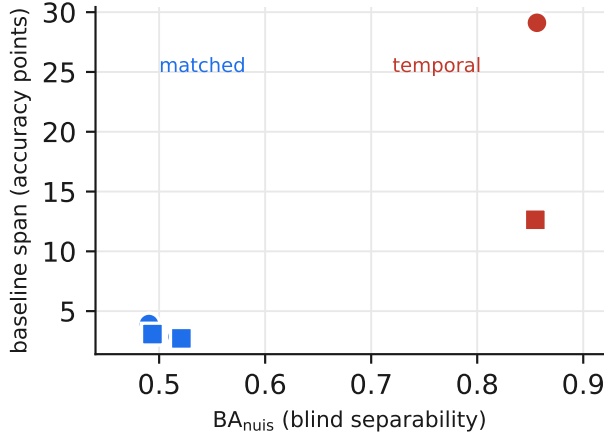


Figure 10: The baseline correction scales with the surface difference it exists to absorb. Circles are the 160M model, squares 410M.

Prefix	words	BA_{nuis}	baseline span	T_{adj}	Outcome
full record	44	0.517	2.6 pts	-0.0040	null, $p = 0.77$
question only	10	0.496	3.0 pts	+0.0015	null, $p = 0.38$

Table 8: Phase 3, measured. Oren et al.’s deliberately contaminated 1.4B checkpoint, PIQA injected at $m = 50$. Both arms are the same test file split at random before training, so Requirement E holds by construction and BA_{nuis} confirms it. The two rows differ only in how much of each record the probe reads. Neither is significant, so the null is not an artefact of a short prefix.

are present in the public file, which is the condition that makes the difference a withheld arm rather than a second population, and the builder refuses if it does not hold.

The measured BA_{nuis} of 0.496 to 0.517 confirms the construction empirically: a classifier with no access to the model cannot separate the injected half from the withheld half. This is the only arm in the paper where that is true, and it is worth contrasting with the 0.855–0.856 of the temporal split.

At $m = 50$ duplications the contrast is null ($T_{\text{adj}} = -0.0040$, $p = 0.77$). We ran it twice, differing only in how much of each record the probe reads, because a null on a ten-word prefix would say more about the prefix than about the instrument. Extending to the full record, which is what the model was trained on, does not change the outcome, and both adjusted statistics are negative.

Two things follow. The null should be read at the sensitivity §4 establishes, which is a real but bounded one, rather than as an absence of any effect. And it extends the calibration point of Table 5, where every published method including an activation probe sits within a few points of chance at $m \in \{1, 2\}$, out to a duplication count fifty times higher.

The baseline profile is again not flat, spanning 2.6 to 3.0 accuracy points, and its shape changes with prefix length (Figure 11): falling on the short prefix, humped with a peak at layer 9 on the full record. Surface separability’s depth profile is a property of the inputs as well as the model, which is a further reason it cannot be assumed.

8.5 What this does not show

The two significant contrasts are on the split where Requirement E fails, and they are therefore uninterpretable as contamination evidence. WikiMIA members are pre-cutoff Wikipedia text and plausibly are in Pythia’s training data, so a true positive is entirely possible; the temporal confound is equally possible; and nothing in the data distinguishes them. That is the point of the gate.

The Phase 3 null is the more informative result, and it should not be overstated either. It is one benchmark, one model, one duplication count, and a null at $m = 50$ does not establish that no familiarity direction exists at any exposure. Converting it into a statement about duplication counts needs the exposure calibration of §10, which requires controlling training. What it does establish is that a linear probe on the residual stream, evaluated against a level-matched baseline on a split where exchangeability is exact, finds nothing on a model that demonstrably memorised the items. Read together with §8, the pattern is that the published positives live on splits that cannot support them, and the splits that can support a verdict return nothing.

9 Reproducibility and Artefacts

Every number in §4 is written into this document by a generated macro file rather than transcribed, so the paper cannot drift from the run that produced it. The released artefacts are:

- `rscp.py`, the protocol, which imports and runs on numpy alone; torch is needed only to extract activations from a real model.
- `simulators.py`, both generative models, with the structural differences of Table 2 computable in one command.
- `phase0c_validation.py`, the study reported here, checkpointed per cell so it can be resumed.
- `test_rscp.py`, 29 property tests covering the claims the method rests on: that the smoother is label-independent and strictly out-of-fold, that its predictions are linear in the labels, that the contrast cancels any constant added to every layer, that the placebo refuses a degenerate surface key rather than returning a value, and that a planted effect of known size is recovered.

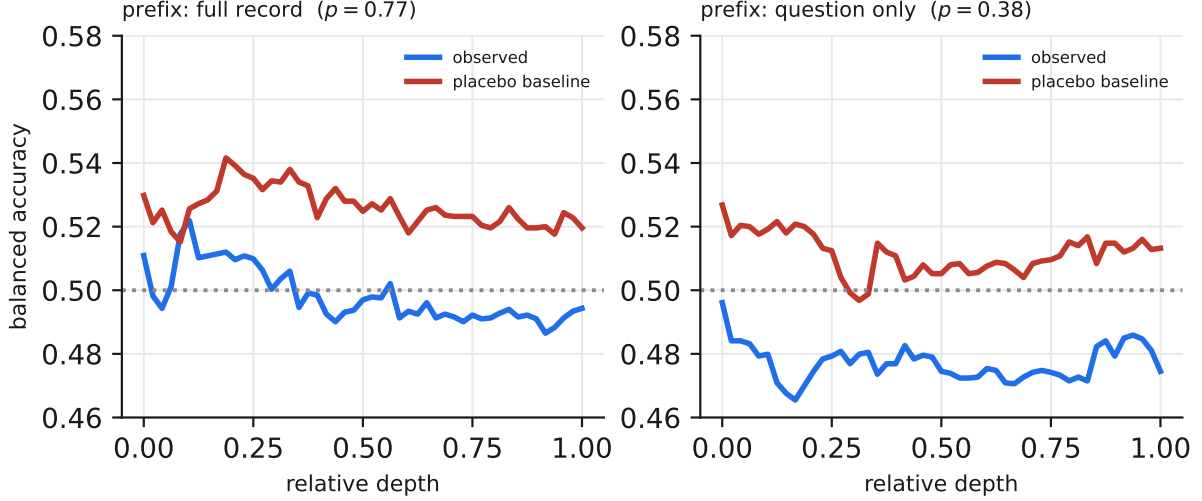


Figure 11: Phase 3, measured. Observed depth profile against the level-matched placebo baseline on Oren et al.’s contaminated 1.4B checkpoint, PIQA injected at $m = 50$, under two prefix lengths. The baseline is not flat in either panel and changes shape with prefix length, which is why it has to be measured rather than assumed. Neither observed profile departs from it significantly.

- `run_rscp_eval.py`, the audit runner, with a dry-run mode that exercises the whole path without a GPU.
- A `Makefile` whose `all` target runs the tests, re-runs the study, regenerates every figure and table, and rebuilds this document. It takes about ten minutes on four CPU cores and needs no model weights.

Two of the bugs fixed during development were found by the dry run rather than by inspection: a degenerate surface key produced a non-finite statistic, and that non-finite value was being compared into a *significant* placebo, which would have silently blocked legitimate audits. Both are now covered by tests. We mention it because a contamination detector that fails closed without saying so is worse than no detector.

10 Follow-on Empirical Validation Programme

This section is the validation plan, not a report of results. Nothing in §5 has been run.

10.1 Pre-registered hypotheses

- H1 Nuisance dominance.** On temporally split membership benchmarks, $BA_{\text{nuis}} \geq 0.65$, and Δ_ℓ is small relative to $BA_{\text{nuis}} - 0.5$. Falsified if $BA_{\text{nuis}} < 0.55$ on WikiMIA-style splits.
- H2 Exposure monotonicity.** Under randomised injection, T increases monotonically in m and is fit by Eq. (11) with $R^2 \geq 0.8$. Falsified by a non-monotone or flat curve.

H3 Layer profile. The depth contrast is positive and ℓ^* falls in the middle-to-late third, not at $\ell \leq 2$. Falsified if ℓ^* concentrates in the first two layers, which would indicate the signal is embedding-level and hence nuisance that Φ_0 failed to capture.

H0 Depth profile of surface separability. On real models the placebo baseline is not flat, and its shape differs across models and item pools. This is the assumption §4 had to remove rather than defend, and measuring the actual shapes is the first thing Phase 1 delivers. Falsified if placebo profiles turn out flat, in which case the recentring is unnecessary machinery.

H4 Extraction-point separation. Full-item extraction yields substantially larger apparent T than prefix extraction on item sets constructed to differ in difficulty but not in membership. Falsified if the two extraction points agree on such sets, which would remove the motivation for the prefix rule.

H5 Recency decay. On Pythia, T decreases with training steps elapsed since exposure. Falsified by a flat profile.

H3 and H4 are the two that carry the paper’s specific methodological claims. H2 determines whether calibration is possible at all.

10.2 Phases

Phase 0, procedure validation. Done; reported in §4. It falsified three of our own design choices, promoted the placebo from a veto to the baseline, and added the $|R| = 2|S|$ requirement.

Phase 1, nuisance audit. Done; reported in §8. It measured real baseline depth profiles for the first time, confirmed the flat-profile assumption is violated, and returned null on every well-matched control.

Phase 1, nuisance audit of existing benchmarks. Compute BA_{nuis} for WikiMIA, for the arXiv-based split of Liu et al. [42], for Pile train versus validation per subdomain, and for GSM8K versus GSM1k. This phase requires forward passes only and produces a standalone result: a table of how much of each existing evaluation split is blind-separable. It also directly extends Das et al. [20] to splits they did not cover.

Phase 2, calibration by injection. Continued pretraining of Pythia-160M and Pythia-410M on a held-out Pile slice with randomised injection at $m \in \{0, 1, 2, 4, 8, 16, 32\}$ across the four format conditions of §5.2. Fit Eq. (11); report m^* and the saturation point. This is the compute-dominant phase.

Phase 3, public contaminated checkpoints. Done, and reported in §8.4: RSCP on the checkpoints of Oren et al. [49] at $m = 50$, the highest duplication count their release provides, directly comparable to the published AUCs in Table 5. The remaining work here is the lower duplication counts and the smaller scales, which their release also covers.

Phase 4, application. GSM8K versus GSM1k across the three models of §5.3, with the OLMo corpus search as partial ground truth.

Phase 5, robustness. Ridge versus logistic probes; (pre, last) versus (pre, mean); sensitivity to n ; the pruned-benchmark question of §6; and the scale trend across the Pythia suite.

10.3 Compute budget

The figures below are estimates derived from parameter counts and token counts, not measurements.

Activation extraction is negligible: a forward pass over 4000 items of roughly 200 tokens on a 1.4B model is on the order of 10^{15} floating-point operations, under a minute on a consumer GPU, with a few hundred megabytes of cache per model. Inference is seconds per configuration, as §4 measured.

Phase 2 dominates. Continued pretraining of Pythia-410M on 2×10^8 tokens costs roughly $6ND \approx 5 \times 10^{17}$ operations; at an effective 3×10^{13} operations per second that is about five hours, and two hours at 160M. Seven exposure levels times four format conditions times two model sizes is a large grid, so we recommend the full grid at 160M and a reduced grid at 410M, for an estimated 60 to 80 GPU-hours, roughly 100 for the programme, a fortnight on one consumer card. Phases 1, 3 and 4 need forward passes only and run on Apple silicon; Phase 2 wants a discrete GPU.

10.4 Artefacts

Release the cached activations for all public-checkpoint configurations, the fitted probe weights and nuisance weights, the permutation null distributions rather than only the p -values, the injection corpora and continued-pretraining checkpoints from Phase 2, and the analysis code with fixed seeds. The permutation nulls matter: they let others recompute a p -value under a different test statistic without rerunning the pipeline. Evaluation should use a standard harness so that accuracy numbers in §3.8 are comparable to published ones [28, 63].

10.5 Decision rule

If H2 holds and $m^* \leq 4$ for the verbatim condition, RSCP is a usable auditing instrument and the follow-on work is deployment across benchmarks with commissioned twins. If H2 holds but m^* is large, the contribution is the detection floor itself, reported as a bound on what activation probing can achieve. If H2 fails, the honest conclusion is that representational familiarity is not a stable function of exposure count at these scales, and Phase 1’s nuisance audit is then the paper’s contribution while the probe is not.

11 Conclusion

Probing the residual stream is a reasonable way to look for benchmark contamination, and the obvious way to do it does not work. Reporting the level of a probe’s advantage over a control makes the false positive rate a function of how large the analyst made the control. Comparing the depth profile against a flat expectation fails whichever way real profiles slope, and they slope both ways. Resampling items with the probe held fixed understates the variance of the statistic. Estimating the baseline on half as many items as the comparison is not a baseline.

What is left after removing those is a protocol with four parts: read activations at the item prefix so the detector cannot see the model’s own answer; contrast the depth profile rather than its level; recentre on a placebo baseline built from the reference set at matched size and matched embedding-layer separability; and test against a permutation null that refits the probe. Its central move is to stop assuming what a null depth profile looks like and measure it per model instead, using data the audit already requires. Under two deliberately dissimilar simulators it holds its nominal error rate where each of the discarded alternatives fails, and it reaches useful power at separations of one to two accuracy points.

Phase 1 has now been run, and it supports the design without settling the science. Real baseline profiles are not flat, they are hump-shaped, and their departure from flatness scales with how different the two item sets are as text. The well-matched negative controls come back null at both scales. The protocol declines to rule on the one split

where a positive appeared, because that split is a temporal one on which a blind classifier already reaches 0.856.

The honest summary of the evidence is therefore narrow. A procedure behaves correctly on synthetic activations where several plausible alternatives do not; it runs on real models; its central assumption was worth removing. Whether a transformer carries a linearly decodable familiarity signal, and at what duplication count, is untouched by any of it, because the only positive result sits behind a failed exchangeability check. Settling that needs the injection experiments of §10 and a benchmark with a reference set that qualifies.

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